

CHE 230: Molecular Systems in Chemical Engineering Homework 2

Due: 2/27/2026 23:59.

CHE 230

1. Of the metals listed in the Table below, (12 points)

- Which will experience the greatest percent reduction in area? Why?
- Which is the strongest? Why?
- Which is the stiffest? Why?
- Which is the hardest? Why?

Material	Yield Strength (MPa)	Tensile Strength (MPa)	Strain at Fracture	Fracture Strength (MPa)	Elastic Modulus (GPa)
A	310	340	0.23	265	210
B	100	120	0.40	105	150
C	415	550	0.15	500	310
D	700	850	0.14	720	210
E	Fractures before yielding			650	350

- B will have the greatest reduction in area as it has the largest strain at fracture.
- D is the strongest as it has the highest tensile strength since it can handle the most stress.
- The stiffest will be C as it has the largest elastic module, meaning more stress is needed for elastic strain.
- The hardest is D as it has the highest yield strength and tensile strength.

2. Consider a cylindrical titanium wire 3.0 mm (0.12 in.) in diameter and $2.5 \cdot 10^4$ mm (1000 in.) long. Calculate its elongation when a load of 500 N (112 lbf) is applied. Assume that the deformation is totally elastic. (8 points)

Stress: $\sigma = \frac{F}{A}$

Strain: $\epsilon = \frac{\Delta L}{L}$

Elastic: $\sigma = E \epsilon$

Can do $\epsilon = \frac{\sigma}{E} \rightarrow \frac{\frac{F/A}{E}}{\epsilon} \rightarrow \Delta L = \frac{F}{E} L \rightarrow \epsilon L = \frac{F L}{A E}$

$\Delta L = \epsilon L$

$\epsilon = \frac{\sigma}{E}$ (Elastic module)

diameter: $d = 3\text{mm} \rightarrow 3 \times 10^{-3}\text{m}$

$E_{Ti} = 107\text{GPa} \rightarrow 107 \times 10^9\text{Pa}$

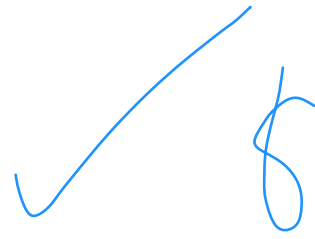
Length: $L = 2.5 \times 10^4\text{mm} \rightarrow 25\text{m}$

load $P = 500\text{N}$

Area $= \frac{\pi}{4} d^2 \rightarrow \frac{\pi}{4} (3 \times 10^{-3})^2 = 7.07 \times 10^{-6}\text{m}^2$

so using

$\Delta L = \frac{FL}{AE} \rightarrow \frac{(500\text{N})(25\text{m})}{(7.07 \times 10^{-6}\text{m}^2)(107 \times 10^9\text{Pa})} = .0165\text{m} \Rightarrow \underline{16.5\text{mm}}$



3. In chemical plants, metallic rods and wires are cold-drawn to improve strength for applications such as heat exchanger tubing, instrumentation lines, and structural supports.

Three candidate materials are being considered:

- Copper
- Brass
- 1040 Steel

Each material is drawn from an initial diameter of 12.0 mm to a final diameter of 10.4 mm.

- Calculate the Percent Cold Work (%CW) produced by the drawing process (2 points).
- Using the provided cold-work diagrams, estimate the following properties at the calculated %CW for:
 - Copper
 - Brass
 - 1040 Steel

For each material, determine:

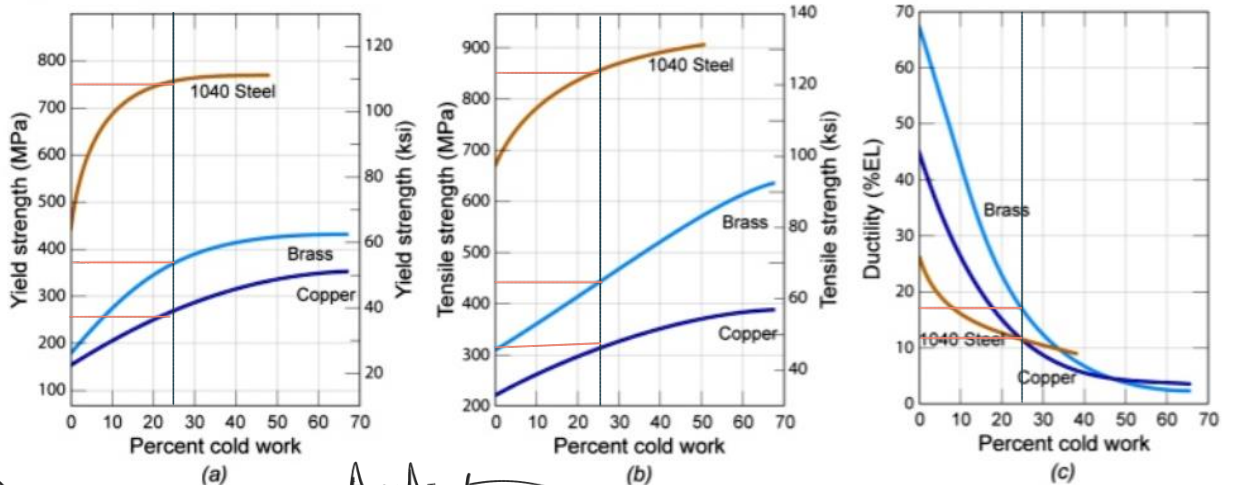
- Yield Strength (MPa)

II. Tensile Strength (MPa)

III. Ductility (% Elongation) (9 points)

c. Based on your results, briefly state:

- Which material is most suitable for high-pressure service
- Which material is best where ductility/formability is important (Answer in 2–3 sentences.) (3 points)



$$d_o = 12 \text{ mm}$$

$$d_f = 10.4 \text{ mm}$$

$$\text{Area: } A = \frac{\pi}{4} d^2$$

a) Cold work: $CW\% = \frac{A_o - A_f}{A_o} \times 100$ ← Cold work area reduction

$$\left(1 - \frac{d_f^2}{d_o^2}\right) \times 100 \rightarrow \left(1 - \frac{10.4^2}{12^2}\right) \times 100 = 24.89\% \approx 25\%$$

b) Copper:

Yield strength: 270 MPa
Tensile strength: 310 MPa
Ductility: 12%

Brass:

Yield strength: 380 MPa
Tensile strength: 440 MPa
Ductility: 18%

1040 Steel:

Yield strength: 750 MPa
Tensile strength: 850 MPa
Ductility: 12%

c) Most suitable for high pressure service is the 1040 steel as it has the highest yield and tensile strength at the same CW%.

The best thing to use where ductility matters is Brass as it has the highest ductility level

4. A single crystal of some metal has a τ_{crss} of 30.7 MPa and is exposed to a tensile stress of 61 MPa.

(i) Will yielding occur when $\phi = 56^\circ$ and $\lambda = 32^\circ$. (3 points)

(ii) If not, what stress is necessary? (3 points)

$$a) \bar{\sigma} = 61 \text{ MPa}$$

Schmid's Law: $\tau_R = \sigma \cos \phi \cos \lambda \rightarrow 61 \text{ MPa} \cos(56^\circ) \cos(32^\circ) = 28.93 \text{ MPa}$

The stress $T_{\text{crss}} > T_R$ so yielding will NOT occur

b) $T_{\text{crss}} = \sigma \underbrace{\cos \phi}_{\text{constant}} \underbrace{\cos \lambda}_{\text{so}}$

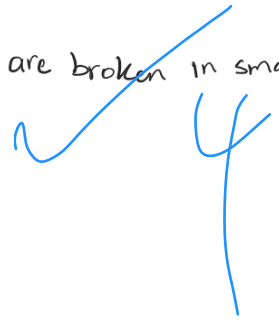
$$30.7 \text{ MPa} = \sigma \cos(56^\circ) \cos(32^\circ) \Rightarrow \sigma = 64.74 \text{ MPa}$$

↑
stress needed ↑

5. a. How are dislocations involved in the plastic deformation of metals/metal alloys? (4 points)

Disslocations are line Defects that appear in metals with plastic like deformation at stresses lower than what's needed to shear.

Slips are dislocations that move alongside a slip plane where bonds are broken in small sections



b. Does the crystal structure of a metal affect its mechanical characteristics? If so, how and why (4 points)?

Yes crystal structures control amount of slip planes and directions exist. By having more slip planes means dislocations can move more easily which can lead to higher ductility yet have easier plastic deformation.



6. a. . How are mechanical properties affected by dislocation mobilities?(4 points)

Dislocations with high mobility have plastic deformations occur at lower stresses meaning it has lower strength and higher ductility.

If dislocations are blocked, it'll have lower mobility which means higher stress is needed to move them. This means it'll have a higher yield strength and hardness but it'll have lower ductility.



b. What techniques are used to increase the strength/hardness of metals/alloys?(4 points)

Some techniques are strength hardening via cold work, solid solution hardening and blocking boundary slips

✓
4

c. How are mechanical characteristics of deformed metal specimens altered by heat treatments? (4 points).

Heat can reduce the cold work via recovery and recrystallization.

✓
4